

AWS Cloud Web Deployment Lab

Project Report



Author

Roberto Almaguer

Role

Software Engineer

Date

June 2026

Executive Summary

This project demonstrates the deployment of a cloud-hosted web application using Amazon Web Services (AWS) and DevOps practices. The objective was to deploy a functional website while applying Linux administration, cloud infrastructure, secure remote access, and deployment automation through GitHub Actions.

Project Objectives

- Deploy a website on AWS.
- Configure an EC2 instance.
- Configure Security Groups.
- Use Linux for server administration.
- Implement GitHub Actions.
- Apply cloud security fundamentals.

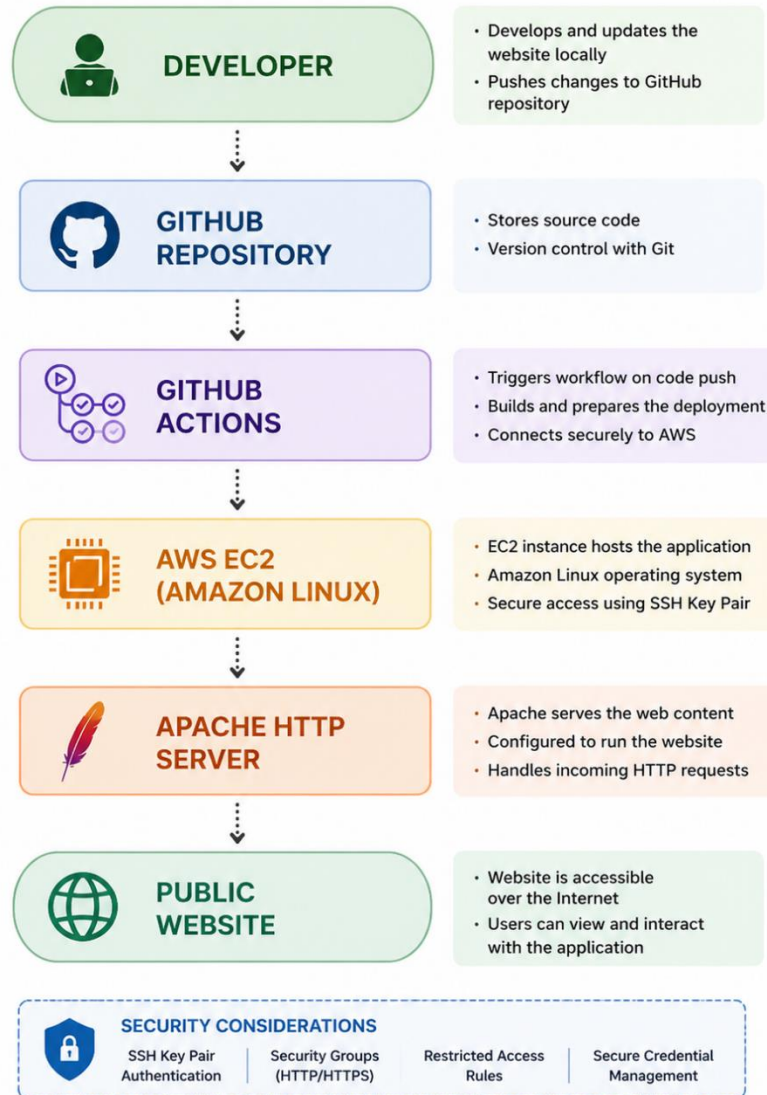
Technologies Used

Technology		 Purpose
	AWS EC2	Virtual server
	Linux	Server administration
	Apache	Web server
	GitHub	Version control
	GitHub Actions	CI/CD
	HTML	Frontend
	CSS	Styling
	JavaScript	Client-side functionality

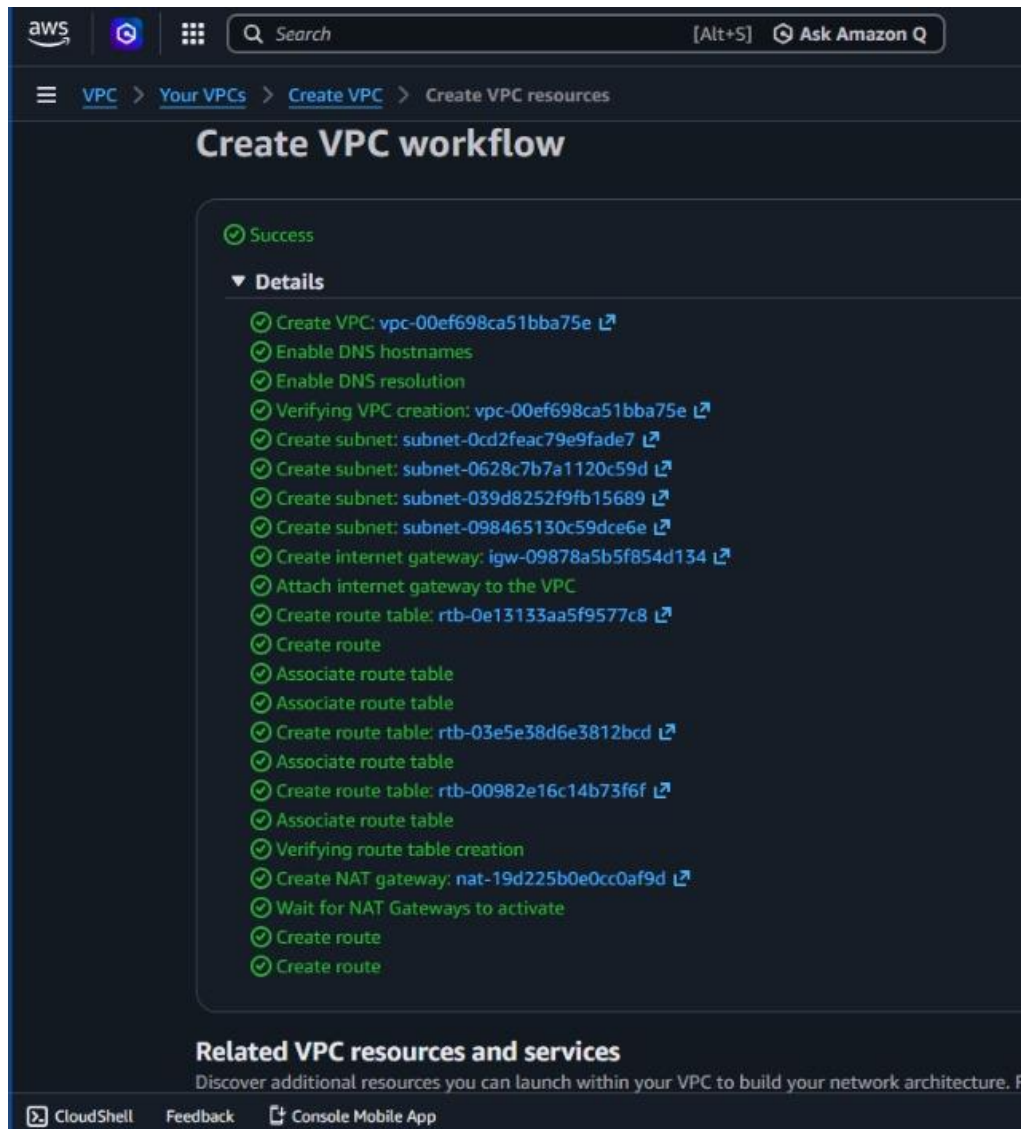
Solution Architecture

SOLUTION ARCHITECTURE

Cloud Web Deployment on AWS using DevOps Practices



Deployment Process



VPC Configuration

A Virtual Private Cloud (VPC) was created to establish an isolated and secure networking environment within Amazon Web Services (AWS). The VPC provides the foundation for hosting cloud resources while allowing full control over network configuration and traffic management.

During this stage, subnets, route tables, an Internet Gateway, and a NAT Gateway were successfully configured, enabling secure communication between the EC2 instance and external users. This network architecture also improves scalability and supports the implementation of security controls for future cloud deployments.

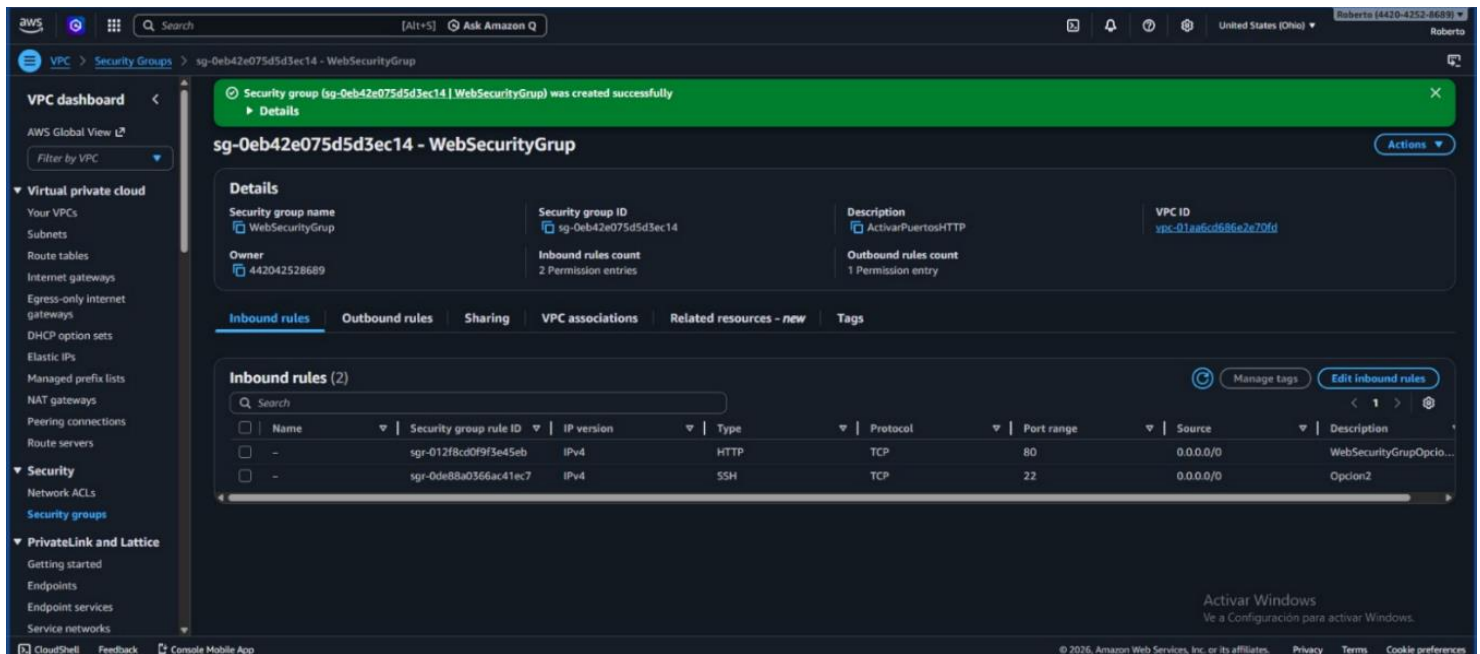
Security Group Configuration

A Security Group was configured to control inbound traffic to the EC2 instance, acting as a virtual firewall for the web server.

The inbound rules were configured to allow only the required services for the application:

- **HTTP (Port 80)** to provide public access to the website.
- **SSH (Port 22)** to enable secure remote administration of the server.

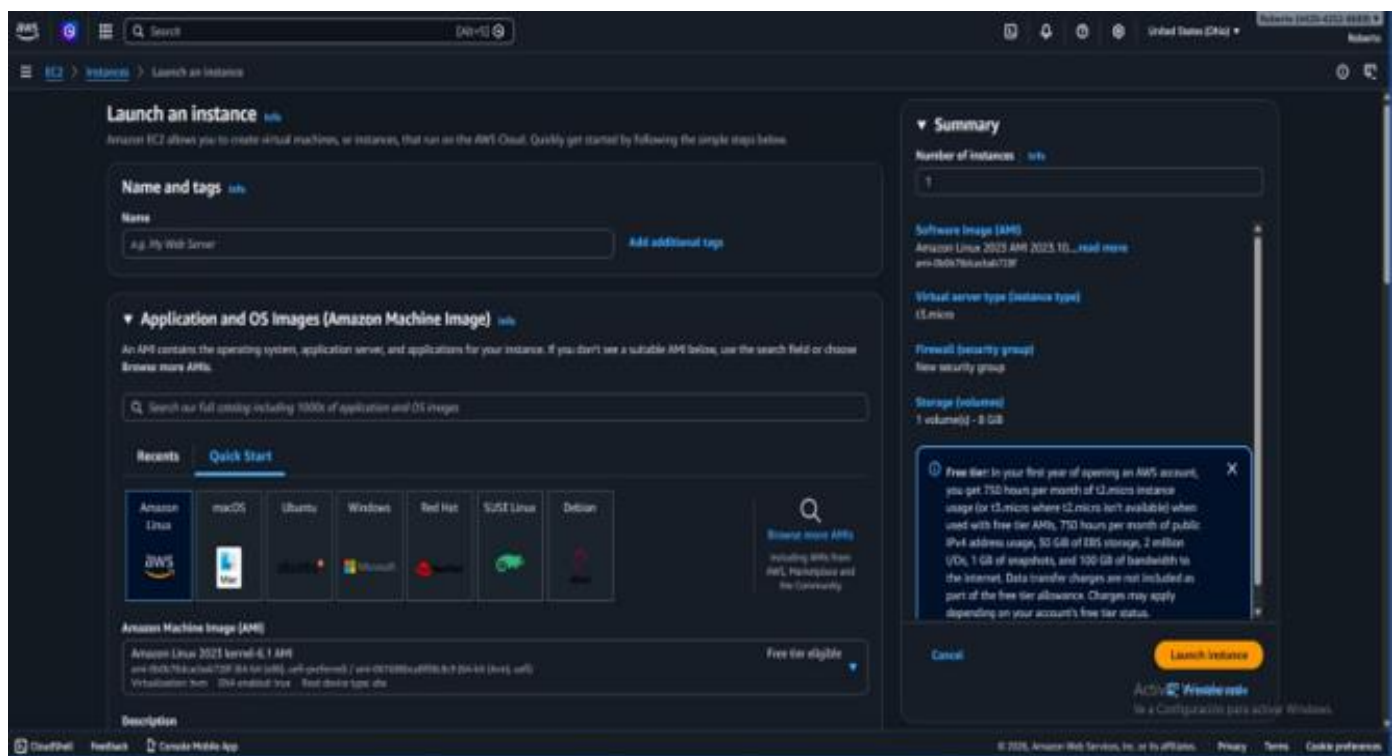
By restricting network access to only the necessary ports, this configuration helps reduce the attack surface and follows the **principle of least privilege**, an essential security best practice in cloud environments.



EC2 Instance Configuration

An Amazon EC2 instance was configured to host the web application in the AWS cloud. During this stage, the Amazon Linux operating system was selected as the server environment, providing a secure and reliable platform for running the Apache web server.

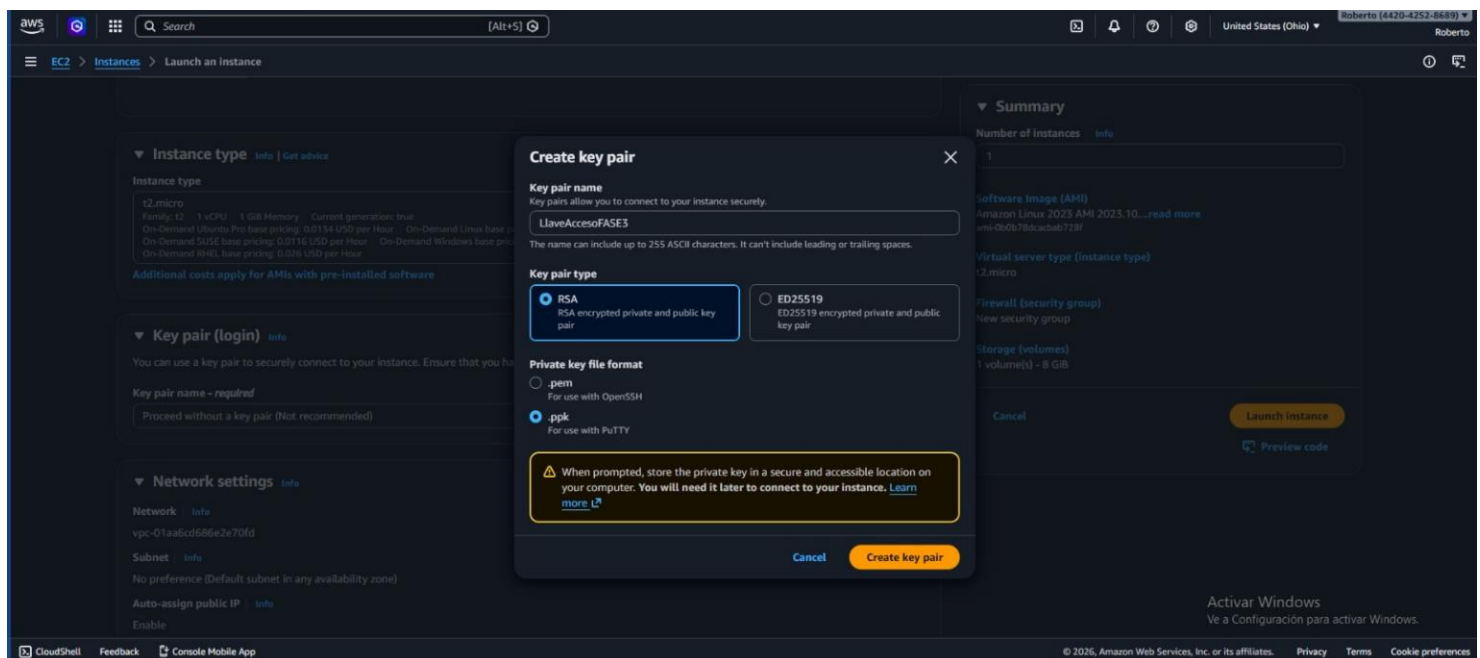
The instance configuration included selecting the appropriate instance type, defining storage capacity, and preparing the server for secure remote administration and web application deployment.



SSH Key Pair Configuration

To enable secure remote access to the EC2 instance, an SSH Key Pair was generated during the deployment process. AWS uses public-key cryptography to authenticate users, eliminating the need for password-based logins and significantly improving server security.

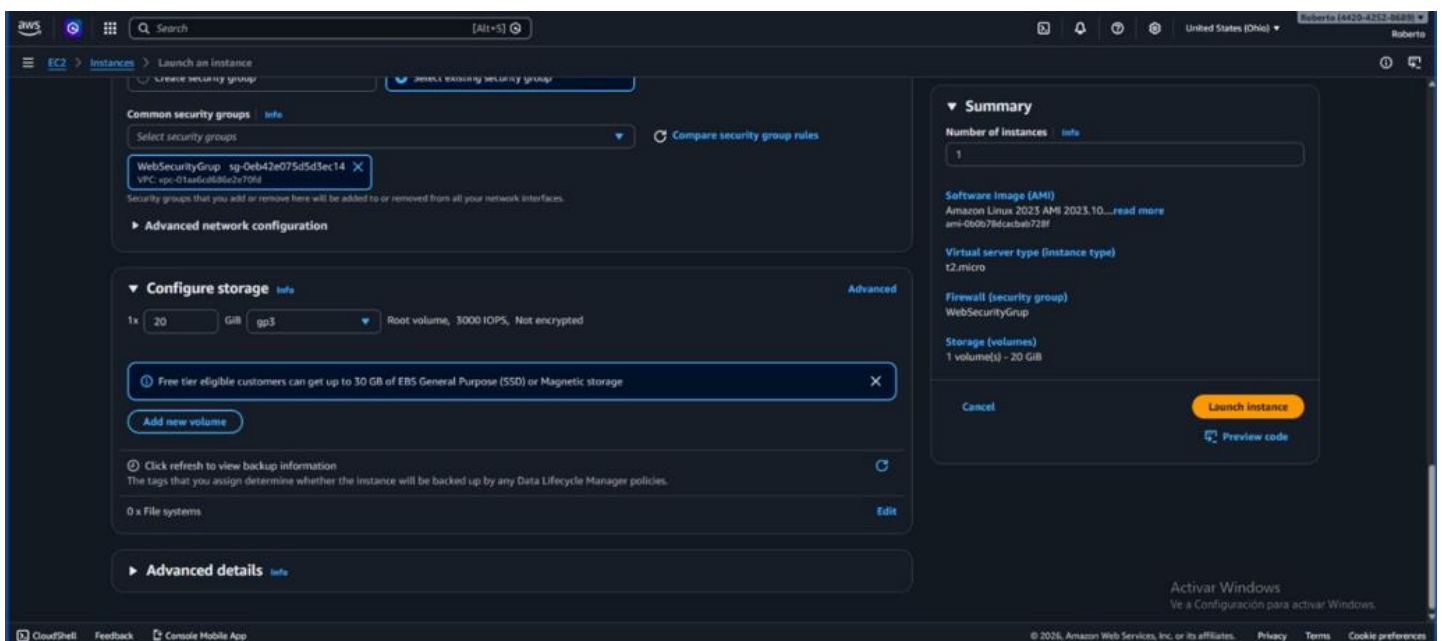
The private key was securely stored and later used to establish encrypted SSH connections to the Linux instance, following AWS security best practices for cloud infrastructure management.



EC2 Deployment Configuration

Before launching the virtual server, the EC2 instance configuration was reviewed to ensure all required components were correctly configured.

This stage included selecting the Amazon Linux operating system, choosing an appropriate instance type, attaching the previously configured Security Group, and allocating storage resources for the web server. Reviewing these settings before deployment helps ensure a secure and stable cloud environment.

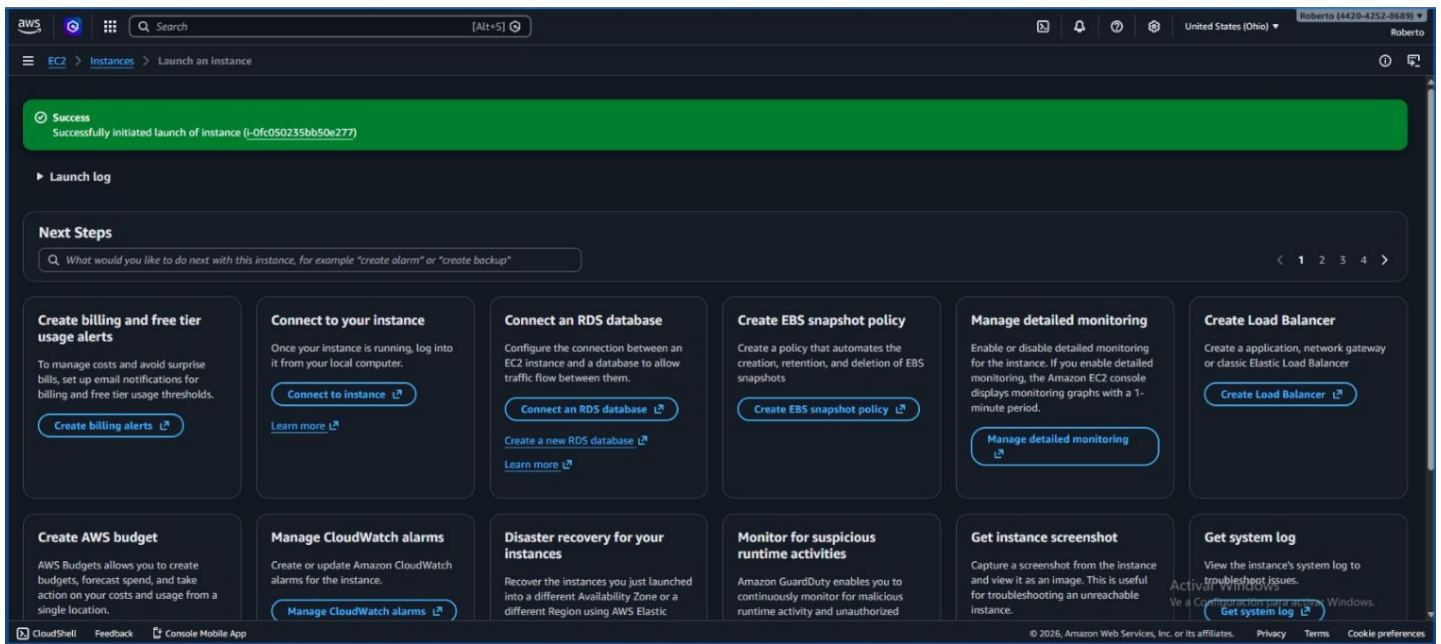


EC2 Instance Deployment

The Amazon EC2 instance was successfully launched after completing the infrastructure configuration. AWS confirmed that the virtual server was deployed

and ready for use, providing the cloud environment required to host the web application.

With the instance running, the server became available for remote administration through SSH and was prepared for the installation and configuration of the Apache HTTP Server, enabling the next stage of the deployment process.



Skills Demonstrated

Skill		Level Demonstrated
	Amazon Web Services (AWS)	✓ Demonstrated
	Amazon VPC	✓ Demonstrated
	Security Groups	✓ Demonstrated
	Amazon EC2	✓ Demonstrated
	Linux Administration	✓ Demonstrated
	Apache HTTP Server	✓ Demonstrated
	Git Version Control	✓ Demonstrated
	GitHub	✓ Demonstrated
	GitHub Actions	✓ Demonstrated
	CI/CD (Continuous Integration / Continuous Deployment)	✓ Demonstrated
	SSH (Secure Shell)	✓ Demonstrated
	Technical Documentation	✓ Demonstrated

Lessons Learned

Throughout this project, I gained practical experience deploying a web application in a cloud environment using Amazon Web Services (AWS).

The project strengthened my understanding of Linux server administration, cloud networking, secure remote access with SSH Key Pairs, and DevOps workflows through GitHub Actions. Additionally, it highlighted the importance of infrastructure planning, security best practices, and deployment automation when managing cloud-hosted applications.

Conclusion

This project successfully demonstrated the deployment of a cloud-hosted web application using Amazon Web Services and DevOps practices.

By integrating cloud infrastructure, Linux administration, GitHub version control, and deployment automation, the project provided valuable hands-on experience with technologies commonly used in modern software development and cloud environments.

Beyond deploying a functional website, this project strengthened my understanding of secure cloud infrastructure, automation, and technical documentation, forming a solid foundation for future work in Cybersecurity, DevOps, and Application Security.

Status

This project was successfully completed as part of my DevOps coursework.

The live AWS deployment is no longer available because the cloud resources were intentionally decommissioned after the project was completed to avoid unnecessary cloud costs.

The source code, technical documentation, architecture, and deployment process remain available in this repository.